Day 1 Introduction to Statistics and Data Analysis

Complete student notes with worked examples and Python practice

Data analysis turns observations into useful evidence. These notes introduce dataset vocabulary, data types, descriptive and inferential statistics, measures of centre and spread, and Python calculations. Each result is explained in terms of its purpose and interpretation.

Introduction to data analysis

Organisations collect data about sales, payments, attendance, customers and website visits. Data becomes useful when it is organised, analysed and interpreted in relation to a clear question.

A statement such as "Sales increased this month" requires context. The comparison period, number of products sold, price changes, branch differences and profit all affect its meaning. One unusually large order may also explain the increase.

A data analyst examines observations, identifies patterns and provides evidence for decisions. Python, SQL, Excel and Power BI support data processing and presentation; statistics helps explain what the results mean.

1 Understanding data before calculations

What statistics means

Statistics involves collecting, organising, summarising, analysing and interpreting data to support decisions. Start with a question, identify the relevant data, and then choose the calculation.

Step	Student performance example
Collect	Collect assessment scores.
Organise	Arrange records by student, batch and module.
Summarise	Calculate average scores and pass percentages.
Analyse	Compare performance across modules.
Interpret	Identify where additional support may be needed.

Read the dataset

This illustrative dataset records practice hours during one week and assessment scores out of 100.

Student ID	Course	Hours	Score	Satisfaction
S01	DA	5	55	Satisfied
S02	PFS	8	70	Very satisfied
S03	DA	3	40	Neutral
S04	PFS	10	85	Satisfied
S05	DA	6	65	Very satisfied

Observation or record: one student’s information. Variable or feature: a recorded characteristic such as score. Value: one entry such as 70. Dataset: the complete collection of records and variables.

Types of data

Type	Meaning	Example
Nominal categorical	Labels without a natural order	Course or city
Ordinal categorical	Ordered categories without known equal gaps	Satisfaction level
Discrete numerical	Counts	Assessments attempted
Continuous numerical	Measurements	Completion time

Student IDs identify records. Their average is not meaningful, even if the IDs contain only digits. A satisfaction order is meaningful, but the gap between categories is not a measured numerical distance.

2 Describing data and drawing conclusions

Descriptive and inferential statistics

Branch	Purpose	Example
Descriptive	Summarise observed data	Describe scores in the current dataset.
Inferential	Use a sample to learn about a population, with uncertainty	Estimate performance across all DA students from a selected sample.

Descriptive statistics answers “What does this dataset look like?” Inferential statistics asks “What can this sample tell us about a wider group?” A descriptive calculation still depends on the quality of the recorded data.

Population and sample

Population: the complete group relevant to the question. Sample: the subset actually examined. If the question concerns all 500 DA students, they form the population. If we examine 50 of them, those 50 form the sample.

A population summary is a parameter; a sample summary is a statistic. For example, the population mean is commonly written as mu, and the sample mean as x-bar.

Selecting only the top-performing 50 students can give a biased picture of a population of 500. Sample size alone does not guarantee representativeness. Selection needs to suit the question and account for relevant groups.

From information to interpretation

An average is a calculation. Explaining what the average tells us, what it misses and how it can guide a decision is analysis. Always state the population, time period, units and comparison group.

Probability as a language for uncertainty

Probability describes how likely an event is, from 0 to 1. An observed proportion is evidence from a particular dataset; it is not automatically a reliable prediction for every future case.

In the example, 3 of 5 students are in DA, so the observed DA proportion is 3/5 = 0.6 or 60%. This describes the five records. It does not establish that 60% of all students at an institute are in DA.

Data quality before analysis

- Identify what one row represents and check for duplicate records.
- Confirm that units, score scales and time periods are consistent.
- Identify missing, invalid or implausible values.
- Check that the dataset covers the group relevant to the question.

3 Mean median and mode

The following numerical dataset illustrates how different measures of central tendency summarise the same observations.

```
values = [12, 15, 15, 18, 20, 22, 22, 22, 25, 90]
```

Mean

The mean adds all values and divides by their count. The total is 261 and there are 10 observations, so the mean is 261 / 10 = 26.1. Every value contributes, so an extreme value can move the mean considerably.

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Median

First sort the values. With an odd count, use the middle value. With an even count, average the two middle values. Here the fifth and sixth values are 20 and 22, so the median is (20 + 22) / 2 = 21.

Mode

The mode is the most frequent value. Here 22 occurs three times, so the mode is 22. A dataset can have multiple modes. Mode can also summarise categories such as the most common course.

Measure	Result	Useful interpretation
Mean	26.1	Uses every value and is sensitive to extremes.
Median	21	Describes the middle position and resists extremes.
Mode	22	Identifies the most frequent value or category.

Effect of an extreme value

If we temporarily exclude 90 for comparison, the remaining total is 171 across 9 values. The mean becomes 19.0. This comparison demonstrates sensitivity; it is not a reason to delete the value.

The mean of 26.1 is greater than every observation except 90. Although it is mathematically correct, it gives an incomplete picture of a typical observation. The median of 21 is closer to the main cluster.

Choosing a suitable summary

- Use the mean when it suits the question and the numerical distribution.
- Consider the median for skewed data or when extremes dominate the mean.
- Use mode for the most common category. Do not average nominal category codes.
- Report context and inspect the distribution rather than relying on one number.

4 Spread consistency and unusual values

The same average can hide different performance

Student	Assessment scores	Mean	Range
A	58, 60, 62	60	4
B	20, 60, 100	60	80

Student A has more consistent observed scores. Student B has much greater variation. The average describes the centre; spread describes how far values lie from that centre.

Measure	Purpose
Range	Maximum minus minimum; sensitive to extremes.
Variance	Average squared deviation for a population; sample estimation uses a correction.
Standard deviation	Square root of variance, expressed in the original units.
Interquartile range	Q3 minus Q1; spread of the middle 50% of observations.

For Student A, treating the three scores as the complete set of interest gives population standard deviation about 1.63 marks. For Student B it is about 32.66 marks. Larger spread here means less consistency, not automatically worse overall ability.

Population and sample formulas

Population variance divides the sum of squared deviations by N. The usual sample variance estimate divides by n - 1. In NumPy, ddof=0 uses the population denominator and ddof=1 uses the sample denominator. Pandas Series.std() defaults to ddof=1. State which version you use.

Investigate outliers

An outlier is unusually distant from other observations. It may be an entry error, a rare valid event or an important finding. Detecting it does not justify automatic deletion.

A common screening rule flags values below Q1 - 1.5 × IQR or above Q3 + 1.5 × IQR. Using NumPy’s default percentile calculation on the PDF example: Q1 = 15.75, Q3 = 22, IQR = 6.25, and fences are 6.375 and 31.375. This flags 90 for investigation.

Quartiles can differ across calculation conventions. Keep the method consistent when comparing results across tools.

Correlation is a relationship rather than proof of cause

Practice hours and scores may move together. A positive Pearson correlation describes an increasing linear association. It does not establish causation. Prior knowledge, attendance and other factors may also influence the scores. Zero Pearson correlation does not rule out a nonlinear relationship.

5 Further statistical topics

Module overview

Topic	Purpose
Probability and distributions	Describe uncertainty and the pattern of possible values.
Sampling and confidence intervals	Estimate population quantities and express uncertainty.
Hypothesis testing	Assess evidence against a stated assumption.
A/B testing	Compare alternatives through a planned experiment.
Linear algebra	Represent and manipulate numerical data using vectors and matrices.

A scalar is one number, a vector is an ordered collection of numbers, and a matrix is a rectangular numerical arrangement. In a typical numerical dataset, rows represent observations and columns represent features.

6 Python approach and dataset inspection

Run the Python examples in order in one Jupyter notebook or Python session. Interpret each output in relation to the question before moving to another calculation.

Environment setup

Install the required packages if they are not already available. Run this command in a terminal. In a notebook, use %pip install numpy pandas matplotlib.

```
python -m pip install numpy pandas matplotlib
```

Step 1 Create the example dataset

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

students = pd.DataFrame({
    "student_id": ["S01", "S02", "S03", "S04", "S05"],
    "course": ["DA", "PFS", "DA", "PFS", "DA"],
    "hours": [5, 8, 3, 10, 6],
    "score": [55, 70, 40, 85, 65],
    "satisfaction": ["Satisfied", "Very satisfied", "Neutral",
                     "Satisfied", "Very satisfied"]
})

print(students)
print("Shape:", students.shape)
print(students.dtypes)
print("Missing values:")
print(students.isna().sum())
```

Expected: shape is (5, 5), meaning 5 rows and 5 columns. Every column has zero missing values. Text dtype names can vary by pandas version; inspect the semantic meaning as well as the storage dtype.

Step 2 Count categories and summarise numerical columns

```
print(students["course"].value_counts())
print(students[["hours", "score"]].describe())
print("Observed DA proportion:",
      (students["course"] == "DA").mean())
```

Expected category counts: DA = 3, PFS = 2. The observed DA proportion is 0.6. describe() gives count, mean, sample standard deviation, minimum, quartiles and maximum for the selected numerical columns.

Variable roles: students stores the table; course is a categorical variable; hours and score are numerical variables. The Boolean comparison is True for DA rows, and its mean gives the proportion of those rows.

7 Python central tendency and interpretation

Step 3 Calculate the central tendency measures

```
values = np.array([12, 15, 15, 18, 20,
                   22, 22, 22, 25, 90])

mean_value = np.mean(values)
median_value = np.median(values)
mode_values = pd.Series(values).mode().tolist()

print("Mean:", mean_value)
print("Median:", median_value)
print("Mode:", mode_values)
print("Sorted values:", np.sort(values))

# Expected output
# Mean: 26.1
# Median: 21.0
# Mode: [22]
```

Variable roles: values holds the numerical observations; mean_value and median_value hold single summaries; mode_values is a list because more than one value may tie for the highest frequency.

Step 4 Verify the result with a manual calculation

```
total = np.sum(values)
count = len(values)
manual_mean = total / count
ordered = np.sort(values)
middle = count // 2
manual_median = (ordered[middle - 1] + ordered[middle]) / 2

print("Total:", total, "Count:", count)
print("Manual mean:", manual_mean)
print("Manual median:", manual_median)
```

Expected: total 261, count 10, mean 26.1 and median 21.0. middle is 5. Python indexing starts at 0, so positions 4 and 5 contain the fifth and sixth values. This median expression is specifically for this even-length dataset.

Step 5 Demonstrate sensitivity to an extreme value

```
# Exclude 90 only for a comparison, retaining the original array.
comparison_values = values[values != 90]
print("Original mean:", np.mean(values))
print("Mean excluding 90:", np.mean(comparison_values))
```

Expected: 26.1 and 19.0. The Boolean mask selects values that differ from 90, while the original values array remains available. The comparison demonstrates sensitivity to an extreme observation. Real observations require investigation before exclusion.

A useful numerical summary can report the median alongside the mean and a measure or visual of spread.

8 Python spread outliers and relationships

Step 6 Compare consistency

```
scores_a = np.array([58, 60, 62])
scores_b = np.array([20, 60, 100])
for name, scores in [("A", scores_a), ("B", scores_b)]:
    print(name, "mean =", np.mean(scores),
          "range =", np.max(scores) - np.min(scores),
          "population std =", round(np.std(scores, ddof=0), 2))
```

Expected: A mean 60.0, range 4, population std 1.63. B mean 60.0, range 80, population std 32.66. scores holds the current student’s array during each loop iteration.

Step 7 Flag unusual values using the IQR rule

```
q1, q3 = np.percentile(values, [25, 75])
iqr = q3 - q1
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr
flagged = values[(values < lower) | (values > upper)]

print("Q1:", q1, "Q3:", q3, "IQR:", iqr)
print("Fences:", lower, upper)
print("Flagged:", flagged.tolist())
```

Expected: Q1 15.75, Q3 22.0, IQR 6.25; fences 6.375 and 31.375; flagged [90]. lower and upper are screening fences, not mandatory deletion thresholds. The | operator combines the two element-wise conditions with OR.

Step 8 Explore a numerical relationship

```
r = students["hours"].corr(students["score"])
print("Pearson correlation:", round(r, 3))
plt.scatter(students["hours"], students["score"])
plt.xlabel("Practice hours this week")
plt.ylabel("Assessment score out of 100")
plt.title("Illustrative practice hours and scores")
plt.show()
```

Expected correlation: about 0.985. The five illustrative points show a strong positive linear association. The tiny constructed dataset does not establish a general relationship or a causal effect. Read the chart alongside the number.

9 Practice exercises and solutions

Exercises

- Explain why student_id should not be averaged.
- Calculate mean and median assessment scores in students.
- Count satisfied and very satisfied students without treating those labels as equally spaced numbers.
- Replace 90 with 190 in a separate copy of values and compare mean and median.
- Use ddof=1 for the consistency example and explain why the standard deviations change.
- Describe one reason why this sample could not support a claim about all students.

Solutions

The student score mean is 63 and the median is 65. Satisfaction counts are Neutral 1, Satisfied 2 and Very satisfied 2. Replacing 90 with 190 changes the numerical example mean to 36.1 while its median remains 21.0. Using sample standard deviation gives A = 2.0 and B = 40.0. The denominator changes from 3 to 2.

Recap questions

- How do data, a calculated summary and an interpretation differ?
- When might a median be more useful than a mean?
- Can two datasets have the same mean and different spread?
- Why must an outlier be investigated?
- What is the difference between an observed association and causation?